

Information Extraction Pipeline

1 Introduction

Information Extraction (IE) is the process of automatically extracting structured information from unstructured text. This structured information can include named entities, relationships, events, and key phrases. The general IE pipeline involves multiple steps, some of which may be necessary for specific tasks but not for others.

2 Text Preprocessing

Before extracting meaningful information, the text must be preprocessed to ensure consistency and improve accuracy.

- **Tokenization:** Splitting text into words or subwords.
- **Lemmatization/Stemming:** Reducing words to their base form (e.g., “running” → “run”).
- **Stopword Removal:** Eliminating common, uninformative words (e.g., “is,” “the”).
- **Sentence Segmentation:** Dividing text into individual sentences.
- **Part-of-Speech (POS) Tagging:** Assigning syntactic categories to words (e.g., noun, verb).
- **Dependency Parsing:** Understanding the syntactic structure of a sentence.

These steps provide the necessary foundation for downstream extraction tasks.

3 Named Entity Recognition (NER)

NER identifies and classifies named entities in the text into predefined categories such as:

- **PERSON** (e.g., “Albert Einstein”)

- **ORGANIZATION** (e.g., “Google”)
- **LOCATION** (e.g., “New York City”)
- **DATE/TIME** (e.g., “March 5, 2023”)
- **MONEY** (e.g., “\$100 million”)

Example:

- **Input:** “Apple Inc. was founded by Steve Jobs in Cupertino in 1976.”
- **NER Output:**
 - **Apple Inc.** → ORG
 - **Steve Jobs** → PERSON
 - **Cupertino** → LOCATION
 - **1976** → DATE

NER typically relies on rule-based, machine learning, or deep learning approaches (e.g., BERT-based NER models).

4 Relation Extraction (RE)

Relation Extraction detects semantic relationships between named entities in text.

Example:

- **Input:** “Barack Obama was born in Hawaii.”
- **Output:** (Barack Obama, born.in, Hawaii)

Techniques for Relation Extraction:

- Rule-based Approaches: Using dependency parsing and pattern matching.
- Supervised Learning: Training classifiers with labeled relation data.
- Deep Learning Models: Using transformers (e.g., BERT, GPT) for relation classification.

5 Coreference Resolution

Coreference Resolution links different mentions of the same entity in a text.

Example:

- **Input:** “Elon Musk founded Tesla. He is also the CEO of SpaceX.”
- **Output:** “He” → Elon Musk

This step is crucial for maintaining context in complex documents.

6 Event Extraction

Event extraction identifies events and their participants.

Example:

- **Input:** “Google acquired YouTube in 2006 for \$1.65 billion.”
- **Output:**
 - **Event Type:** Acquisition
 - **Company Acquired:** YouTube
 - **Acquiring Company:** Google
 - **Date:** 2006
 - **Amount:** \$1.65 billion

7 Key Phrase Extraction (KPE)

Key phrase extraction (KPE) aims to automatically extract important words or phrases summarizing a document.

Mathematical Formulation: Given an input document represented as a sequence of words $X = \{x_1, x_2, \dots, x_T\}$, key phrase extraction can be framed as a sequence labeling problem, where each word x_t receives a binary label y_t .

$$L = - \sum_{i=1}^N \sum_{t=1}^T y_t \log P(y_t|X, \theta) + (1 - y_t) \log(1 - P(y_t|X, \theta))$$

Deep Learning Architectures for KPE:

- TF-IDF and Statistical Methods
- Graph-based Methods (TextRank, PositionRank)
- Supervised Neural Models:
 - BiLSTM-CRF: Uses bidirectional LSTMs with a CRF layer for structured phrase extraction.
 - BERT-based Span Prediction: Fine-tuning BERT for extracting key phrases.
 - Seq2Seq Models: Treats key phrase generation as a text summarization task.

8 Template Filling / Knowledge Graph Construction

Extracted information is structured into templates or stored in a knowledge graph.

Example:

Entity 1	Relation	Entity 2	Date
Google	founded_by	Larry Page	1998
Google	acquired	YouTube	2006

9 Evaluation of IE Systems

IE tasks are evaluated using:

- **Precision:** Correctly extracted information / Total extracted.
- **Recall:** Correctly extracted information / Total relevant.
- **F1-Score:** Harmonic mean of precision and recall.

10 Challenges in Information Extraction

- **Ambiguity:** Words may have multiple meanings (e.g., “Apple” as a company or a fruit).
- **Data Sparsity:** Insufficient labeled data for certain domains.
- **Complex Sentence Structures:** Long-range dependencies affect extraction.
- **Error Propagation:** Mistakes in earlier steps impact later stages.

11 Summary of IE Pipeline

1. Preprocessing → Tokenization, POS tagging, lemmatization.
2. NER → Identify named entities.
3. Relation Extraction → Detect relationships between entities.
4. Coreference Resolution → Link entity mentions.
5. Event Extraction → Identify events and participants.
6. Key Phrase Extraction → Extract main topics.
7. Template Filling → Store structured knowledge.

8. Evaluation $\rightarrow$ Precision, recall, F1-score.
9. Deep Learning Models $\rightarrow$ Transformers, BiLSTMs, CNNs, GNNs.